

Towards a fuzzy logic for automated multi-issue negotiation

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The Problem

- Suppose we have a buyer and a seller (agents)
 - ▶ the buyer describes what he is intended to buy and his preferences
 - ▶ the seller describes what he is intended to sell and his preferences
 - ▶ there is some background knowledge
- The objective is determine “an optimal” (**Pareto optimal**) agreement among the two

Our Approach

- We present a novel logic-based approach
 - ▶ we use a logical language to express the seller's, the buyer's preferences and the background knowledge
- In particular, we introduce a **mathematical fuzzy propositional logic** extended with **concrete domains** in order
 - ▶ to handle numerical features
 - ▶ to handle non numerical features
 - ▶ to deal with vagueness in buyer/seller preferences

Running Example

- A car seller sells a sedan car
- A buyer is looking for a second hand passenger car
- Both the buyer as well as the seller have preferences (restrictions)
- Our aim is to find an optimal agreement

Running Example: the Background Knowledge

- 1 A sedan is a passenger car
- 2 A satellite alarm system is an alarm system
- 3 The navigator pack is a satellite alarm system with a GPS system
- 4 The Insurance Plus package is a driver insurance together with a theft insurance
- 5 The car colours are black or grey



Running Example: Buyer's preferences

- 1 He does not want to pay more than 26000 euro (buyer reservation value)
- 2 If there is an alarm system in the car then he is completely satisfied with paying no more than 22300 euro, but he can go up to 22750 euro to a lesser degree of satisfaction
- 3 He wants a driver insurance and either a theft insurance or a fire insurance
- 4 He wants air conditioning and the external colour should be either black or grey
- 5 Preferably the price is no more than 22000 euro, but he can go up to 24000 euro to a lesser degree of satisfaction
- 6 The kilometer warranty is preferably at least 175000, but he may go down to 150000 to a lesser degree of satisfaction
- 7 The weights of the preferences 2-6 are, $(0.1, 0.2, 0.1, 0.2, 0.4)$. The higher the value the more important is the preference



Running Example: Seller's preferences

- 1 He wants to sell no less than 22000 euro (seller reservation value)
- 2 If there is an navigator pack system in the car then he is completely satisfied with selling no less than 22750 euro, but he can go down to 22500 euro to a lesser degree of satisfaction
- 3 Preferably the seller sells the Insurance Plus package
- 4 The kilometer warranty is preferably at most 100000, but he may go up to 125000 to a lesser degree of satisfaction
- 5 The monthly warranty is preferably at most 60, but he may go up to 72 to a lesser degree of satisfaction
- 6 If the color is black then the car has air conditioning
- 7 The weights of the preferences 2-6 are, (0.3, 0.1, 0.3, 0.1, 0.2). The higher the value the more important is the preference

The Language $\mathcal{P}(\mathcal{N})$: Informally

We consider

- propositional atoms $A_1, A_2, \dots$ from a finite set $\mathcal{A}$
 - ▶ *e.g.*, Sedan, DriverInsurance, ...
- numerical features $f_1, f_2, \dots$, each one taking values in its specific domain $D_{f_1}, D_{f_2}, \dots$,
 - ▶ such as month_warranty with domain $[0, 96]$ (months), or price with domain $[1.000, 50.000]$ (euros)
- we may write, *e.g.*, $(\text{price} \leq 20.000)$ and $(\text{month_warranty} \geq 60)$, to constrain the value of the features
- combination of constraints forming complex propositional requirements
 - ▶ *e.g.*, $\text{Sedan} \wedge (\text{price} \leq 25.000) \wedge (\text{deliverytime} \leq 30)$



We consider also

- **fuzzy constraints**

- ▶ numerical features are constrained by so-called fuzzy membership functions

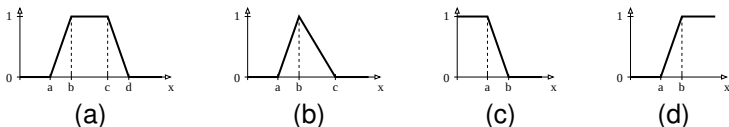


Figure: (a) Trapezoidal function $trz(a, b, c, d)$, (b) triangular function $tri(a, b, c)$, (c) left shoulder function $ls(a, b)$, and (d) right shoulder function $rs(a, b)$.

- ▶ For instance, $(price \text{ } ls(18000, 22000))$ dictates that given a price it returns the degree to which the constraint is satisfied

The Language $\mathcal{P}(\mathcal{N})$

Definition (Concrete Domain)

A **concrete domain** D consists of a non-empty set $\Delta_c(D)$ of numerical values, and a set of predicates $C(D)$ expressing numerical constraints on D .

- predicates will always be of the form
 - ▶ $\geq k, \leq k, = k$ (k is a value in D), e.g., (`price` $\leq$ 26000), called **hard constraint**
 - ▶ $ls(a, b), rs(a, b), tri(a, b, c), trz(a, b, c, d)$, e.g., (`deliverytime` $ls(30, 40)$), called **soft constraint**

Definition (The language $\mathcal{P}(\mathcal{N})$)

Let $\mathcal{A}$ be a set of propositional atoms, and F a set of pairs $\langle f, D_f \rangle$ each made of a feature name and an associated concrete domain D_f , and let k be a value in D_f . Then the following formulas are in $\mathcal{P}(\mathcal{N})$:

- 1 every atom $A \in \mathcal{A}$ is a formula in $\mathcal{P}(\mathcal{N})$
- 2 if $\langle f, D_f \rangle \in F$, $k \in D_f$, and $c \in \{\geq, \leq, =\}$ then $(f \ c \ k)$ is a formula in $\mathcal{P}(\mathcal{N})$
- 3 if $\langle f, D_f \rangle \in F$ and c is of the form $ls(a, b), rs(a, b), tri(a, b, c), trz(a, b, c, d)$ then $(f \ c)$ is a formula in $\mathcal{P}(\mathcal{N})$
- 4 if ψ and φ are formulas in $\mathcal{P}(\mathcal{N})$ and $n \in [0, 1]$ then so are $\neg\psi, \psi \wedge \varphi, \psi \vee \varphi, \psi \Rightarrow \varphi$ and $n \cdot \psi$. We use $\psi \Leftrightarrow \varphi$ in place of $(\psi \Rightarrow \varphi) \wedge (\varphi \Rightarrow \psi)$.

The Language $\mathcal{P}(\mathcal{N})$: semantics

- We need to give meaning to atomic constructs and logical connectives
- As an interpretation function $\mathcal{I}$ maps propositional atoms into $[0, 1]$, feature names into values in their domain, connectives are functions into $[0, 1]$, *i.e.*,
 - ▶ $\neg: [0, 1] \rightarrow [0, 1]$
 - ▶ $\wedge: [0, 1] \times [0, 1] \rightarrow [0, 1]$, so-called **t-norm**
 - ▶ $\vee: [0, 1] \times [0, 1] \rightarrow [0, 1]$, so-called **s-norm**
 - ▶ $\Rightarrow: [0, 1] \times [0, 1] \rightarrow [0, 1]$
- The choice of them is not arbitrary, but is usually restricted: *e.g.*,

	Łukasiewicz Logic	Gödel Logic	Product Logic	Zadeh
$\neg x$	$1 - x$	if $x = 0$ then 1 else 0	if $x = 0$ then 1 else 0	$1 - x$
$x \wedge y$	$\max(x + y - 1, 0)$	$\min(x, y)$	$x \cdot y$	$\min(x, y)$
$x \vee y$	$\min(x + y, 1)$	$\max(x, y)$	$x + y - x \cdot y$	$\max(x, y)$
$x \Rightarrow y$	if $x \leq y$ then 1 else $1 - x + y$	if $x \leq y$ then 1 else y	if $x \leq y$ then 1 else y/x	$\max(1 - x, y)$

Property	Łukasiewicz Logic	Gödel Logic	Product Logic	Zadeh Logic
$x \wedge \neg x = 0$	•	•	•	
$x \vee \neg x = 1$	•			
$x \wedge x = x$		•		•
$x \vee x = x$		•		•
$\neg \neg x = x$	•			•
$x \Rightarrow y = \neg x \vee y$	•			•
$\neg(x \Rightarrow y) = x \wedge \neg y$	•			•
$\neg(x \wedge y) = \neg x \vee \neg y$	•	•	•	•
$\neg(x \vee y) = \neg x \wedge \neg y$	•	•	•	•

Definition (Interpretation and models)

An interpretation $\mathcal{I}$ for $\mathcal{P}(\mathcal{N})$ is a function (denoted as a superscript $\cdot^{\mathcal{I}}$ on its argument) that maps each atom in $\mathcal{A}$ into a truth value $A^{\mathcal{I}} \in [0, 1]$, each feature name f into a value $f^{\mathcal{I}} \in D_f$, and assigns truth values in $[0, 1]$ to formulae as follows:

- for hard constraints, $(f \text{ c } k)^{\mathcal{I}} = 1$ iff the relation $f^{\mathcal{I}} \text{ c } k$ is true in D_f , $(f \text{ c } k)^{\mathcal{I}} = 0$ otherwise
- for soft constraints, $(f \text{ c })^{\mathcal{I}} = c(f^{\mathcal{I}})$, i.e., the result of evaluating the fuzzy membership function c on the value $f^{\mathcal{I}}$
- $(\neg \psi)^{\mathcal{I}} = \neg \psi^{\mathcal{I}}$, $(\psi \wedge \varphi)^{\mathcal{I}} = \psi^{\mathcal{I}} \wedge \varphi^{\mathcal{I}}$, $(\psi \vee \varphi)^{\mathcal{I}} = \psi^{\mathcal{I}} \vee \varphi^{\mathcal{I}}$, $(n \cdot \psi)^{\mathcal{I}} = n \cdot \psi^{\mathcal{I}}$ and $(\psi \Rightarrow \varphi)^{\mathcal{I}} = \psi^{\mathcal{I}} \Rightarrow \varphi^{\mathcal{I}}$.

Given a formula φ in $\mathcal{P}(\mathcal{N})$, we denote with $\mathcal{I} \models \varphi$ the fact that $\mathcal{I}$ assigns 1 to φ . If $\mathcal{I} \models \varphi$ we say $\mathcal{I}$ is a *model* for φ , and $\mathcal{I}$ is a model for a set of formulas when it is a model for each formula.

Intended semantics

- We use Łukasiewicz Logic as the specific interpretation of the connectives
- The reason for this choice:
 - ▶ nice logical and computational properties (e.g., , NP-completeness, Pareto optimality)
 - ▶ $x \wedge_G y = \min(x, y) = x \wedge (x \Rightarrow y)$
 - ▶ $x \vee_G y = \max(x, y) = \neg(\neg x \wedge_G \neg y)$
- Main inference task: determine the **maximal truth degree** of ϕ over all models of a theory $\mathcal{T}$:

$$\langle \phi \rangle_{\mathcal{T}} = \sup_{\mathcal{I} \models \mathcal{T}} \phi^{\mathcal{I}}$$

Problem Encoding

- Background knowledge: set $\mathcal{T}$ of $\mathcal{P}(\mathcal{N})$ formulae

$$\mathcal{T} = \left\{ \begin{array}{l} \text{Sedan} \Rightarrow \text{PassengerCar} \\ \text{ExternalColorBlack} \Rightarrow \neg \text{ExternalColorGray} \\ \text{SatelliteAlarm} \Rightarrow \text{AlarmSystem} \\ \text{InsurancePlus} \Leftrightarrow \text{DriverInsurance} \wedge \text{TheftInsurance} \\ \text{NavigatorPack} \Leftrightarrow \text{SatelliteAlarm} \wedge \text{GPS_system} \end{array} \right.$$

Hard Constraints

- The seller and the buyer model with formulae σ and β the minimal requirements they accept for the negotiation
 - ▶ $\mathcal{T} \cup \{\beta\}$ and $\mathcal{T} \cup \{\sigma\}$ have to be satisfiable
 - ▶ **possible agreement**: interpretation $\mathcal{I}$ between β and σ such that $\mathcal{I} \models \mathcal{T} \cup \{\sigma, \beta\}$
 - ▶ Note: if $\mathcal{T} \cup \{\sigma, \beta\}$ has no models, then the negotiation ends immediately
- E.g., $\beta = \text{PassengerCar} \wedge (\text{price} \leq 26000)$
- E.g., $\sigma = \text{Sedan} \wedge \text{price} \geq 24000$

Soft Constraints

- The seller and the buyer model preferences with formulae β_i and σ_i
- The buyer's **negotiation preference** $\mathcal{B}$ is formula of the form $n_1 \cdot \beta_1 \vee \dots \vee n_k \cdot \beta_k$ (with $n_i \in [0, 1]$, $\sum_i n_i = 1$)
- The seller's **negotiation preference** $\mathcal{S}$ is formula of the form $m_1 \cdot \sigma_1 \vee \dots \vee m_h \cdot \sigma_h$ (with $m_i \in [0, 1]$, $\sum_i m_i = 1$)
 - ▶ Note that under Łukasiewicz logic, $n_1 \cdot \beta_1 \vee \dots \vee n_k \cdot \beta_k$ is a weighted sum of the degree of truth of the β_i

Example: Buyer's Soft Constraints

$$\beta_1 = \text{AlarmSystem} \wedge (\text{price}, \text{rs}(23000, 26000))$$

$$\beta_2 = \text{DriverInsurance} \wedge (\text{TheftInsurance} \vee \text{FireInsurance})$$

$$\beta_3 = \text{AirConditioning} \wedge (\text{ExternalColorBlack} \vee \text{ExternalColorGray})$$

$$\beta_4 = (\text{price}, \text{ls}(22000, 24000)) \vee (\text{km_warranty}, \text{rs}(140000, 160000))$$

$$B = 0.1 \cdot \beta_1 \vee 0.2 \cdot \beta_2 \vee 0.3 \cdot \beta_3 \vee 0.4 \cdot \beta_4$$

Example: Seller's Soft Constraints

$\sigma_1 = \text{NavigatorPack} \wedge (\text{price}, \text{rs}(24000, 26000))$

$\sigma_2 = \text{InsurancePlus}$

$\sigma_3 = (\text{km_warranty}, \text{ls}(150000, 170000))$

$\sigma_4 = \text{ExternalColorBlack} \wedge \text{AirConditioning}$

$S = 0.3 \cdot \sigma_1 \vee 0.1 \cdot \sigma_2 \vee 0.4 \cdot \sigma_3 \vee 0.2 \cdot \sigma_4$

Pareto Agreements

- Given background theory $\mathcal{T}$
- Buyer's hard constraint β
- Buyer's negotiation preference $\mathcal{B} := n_1 \cdot \beta_1 \vee \dots \vee n_k \cdot \beta_k$
- Seller's hard constraint σ
- Seller's negotiation preference $\mathcal{S} := m_1 \cdot \sigma_1 \vee \dots \vee m_h \cdot \sigma_h$
- **Pareto agreement value $\bar{p}$:**

$$\bar{p} = \langle \mathcal{B} \wedge \mathcal{S} \rangle_{\mathcal{T} \cup \{\beta, \sigma\}}$$

- **Pareto agreement:** interpretation $\bar{\mathcal{I}}$ such that

$$\bar{\mathcal{I}} = \arg \max_{\mathcal{I} \models \mathcal{T} \cup \{\beta, \sigma\}} \mathcal{B}^{\mathcal{I}} \wedge \mathcal{S}^{\mathcal{I}}$$

- Note that:
 - ▶ Pareto agreement value $\bar{p}$ is unique
 - ▶ For any Pareto agreement $\bar{\mathcal{I}}$, $\bar{\mathcal{I}} \models \mathcal{T} \cup \{\beta, \sigma\}$ and $\bar{p} = \mathcal{B}^{\bar{\mathcal{I}}} \wedge \mathcal{S}^{\bar{\mathcal{I}}}$
 - ▶ Pareto agreement $\bar{\mathcal{I}}$ is not unique

Running Example

- Pareto Agreement value: $\bar{p} := 0.633$
- Pareto Agreement: $\bar{\mathcal{I}}$

Sedan $\bar{\mathcal{I}} = 1.0$,
PassengerCar $\bar{\mathcal{I}} = 1.0$,
InsurancePlus $\bar{\mathcal{I}} = 1.0$,
AlarmSystem $\bar{\mathcal{I}} = 1.0$,
DriverInsurance $\bar{\mathcal{I}} = 1.0$,
AirConditioning $\bar{\mathcal{I}} = 1.0$,
NavigatorPack $\bar{\mathcal{I}} = 1.0$,
(km_warranty_{ls}(150000,170000)) $\bar{\mathcal{I}} = 0.5$, i.e., km_warranty $\bar{\mathcal{I}} = 160000$,
(price_{ls}(23000,26000)) $\bar{\mathcal{I}} = 0.33$, i.e., price $\bar{\mathcal{I}} = 24000$,
TheftInsurance $\bar{\mathcal{I}} = 1.0$,
FireInsurance $\bar{\mathcal{I}} = 1.0$,
ExternalColorBlack $\bar{\mathcal{I}} = 1.0$,
ExternalColorGray $\bar{\mathcal{I}} = 0.0$.

Computing Optimal Agreements

- Can we compute the Pareto agreements (i.e., $\bar{p}$ and $\bar{l}$)? Yes (NP-complete)
- Solution: combination of logical inference (like Pavelka's Rational Logic) + Mixed Integer Linear Programming
- For a rational $r \in [0, 1]$,
 - ▶ a formula $r \Rightarrow \phi$ means "The degree of truth of ϕ is greater or equal than r "
 - ▶ a formula $\phi \Rightarrow r$ means "The degree of truth of ϕ is smaller or equal than r "
- Let $\bar{\mathcal{T}} = \{1 \Rightarrow \phi \mid \phi \in \mathcal{T} \cup \{\beta, \sigma\}\}$
- Apply logical inference rules to $\bar{\mathcal{T}} \cup \{x \Rightarrow \mathcal{B} \wedge \mathcal{S}\}$, obtaining a set of in-equations $IE(\mathcal{T}, \beta, \sigma, \mathcal{B}, \mathcal{S}, x)$
 - ▶ Logical inference rules (excerpt):
- Solve Mixed Integer Linear Programming problem

$$\max x. \ IE(\mathcal{T}, \beta, \sigma, \mathcal{B}, \mathcal{S}, x)$$

- Any solution is a Pareto agreement

- ▶ Indeed:

$$\bar{p} = \langle \mathcal{B} \wedge \mathcal{S} \rangle_{\mathcal{T} \cup \{\beta, \sigma\}} = \max x. \ IE(\mathcal{T}, \beta, \sigma, \mathcal{B}, \mathcal{S}, x)$$

Conclusions and future work

- We have presented a logic-based approach for finding Pareto optimal agreements among Agents: it allows
 - ▶ to handle numerical features
 - ▶ to handle non numerical features
 - ▶ to deal with vagueness in agents preferences
- The method has been implemented in much richer languages, such as fuzzy Logic Programming and fuzzy Description Logics
- Future work:
 - ▶ experimentation
 - ▶ investigating other negotiation protocols, without the presence of a mediator, allowing to reach an agreement in a reasonable amount of communication rounds

THANKS :-)